

Diabetes Check: A Multiclass Machine Learning Approach for Early Clinical Screening

Data Mining & Machine Learning Project

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PART 1

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PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

Problem Definition & Dataset

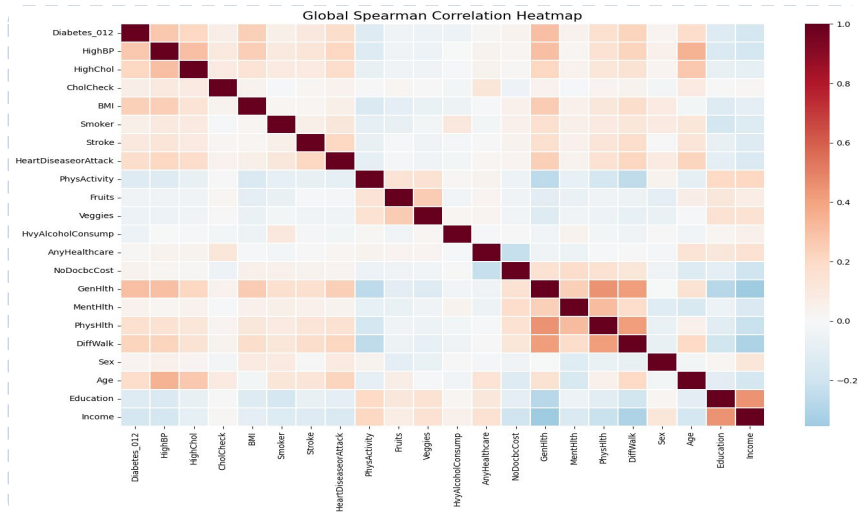
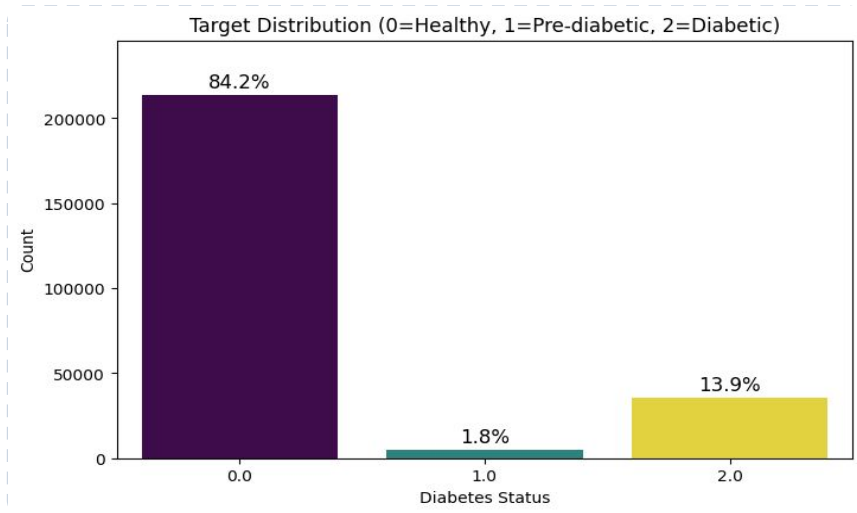
The Problem

- Multiclass classification task: categorize individuals as Healthy (0), Pre-Diabetic (1), or Diabetic (2) based on behavioral and biometric survey data.
- Clinical rationale: preserving three-class granularity enables risk stratification and early preventive intervention.
- Key challenge: The severe class imbalance makes minority-class detection very difficult. In particular, the Pre-Diabetes class, at just 1.8%, is highly prone to misclassification.
- Practical relevance: a non-invasive, low-cost screening tool deployable at first clinical contact, without lab tests.

Dataset at a Glance

Task	<i>Multiclass Classification (3 classes)</i>
Dataset	<i>CDC BRFSS 2015 — 253,680 samples, 22 features</i>
Target	<i>Diabetes_012 (0=Healthy, 1=Pre-Diabetic, 2=Diabetic)</i>
Features	<i>22 binary / ordinal features (all categorical)</i>
Missing values	<i>0% missing values</i>
Class balance	<i>84.2% / 1.8% / 13.9% — highly imbalanced</i>
Split	<i>Nested Stratified Group 10-Fold CV (inner: 5-Fold)</i>
Metrics	<i>Balanced Accuracy (primary), F1-macro, Precision-Recall</i>

Exploratory Data Analysis



Key findings from EDA:

Severe class imbalance: 84.2% Healthy, 13.9% Diabetic, only 1.8% Pre-Diabetic, making this minority class the hardest to detect.

No critical multicollinearity (Spearman $\rho < 0.8$ for all pairs): all 21 predictive features retained for modeling.

23,899 exact duplicate rows found; 1,834 ambiguous profiles span multiple classes (2.41% of dataset). However, we chose to retain them to reflect real-world clinical overlap.

Feature Engineering & Selection

Engineering

- No new features engineered: dataset features are already binary/ordinal survey items
- Standardization (StandardScaler) applied inside CV folds for Logistic Regression
- No imputation needed: zero missing values across all 21 predictive features

Selection Method

- Filter method: Global Spearman correlation heatmap
- Criterion: no pair exceeded $\rho = 0.8/0.9$ threshold
- Validated visually via heatmap; no CV-based selection needed
- Final predictive features retained: all 21

Why These Features?

- All features map to established clinical/epidemiological risk factors (BMI, BP, Cholesterol, GenHlth, Age)
- No features discarded: absence of multicollinearity confirms each carries unique information
- Socioeconomic features (Income, Education, NoDocbcCost) retained as literature-validated social determinants

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

- All preprocessing fitted only on training folds
- sklearn Pipeline + ColumnTransformer: StandardScaler fitted on train fold, applied to val/test. No information from validation or test sets leaked during training

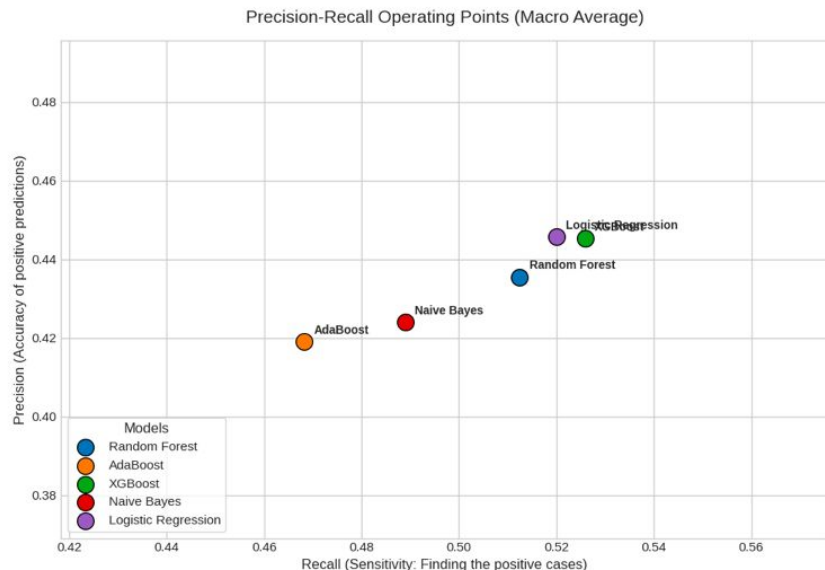
Class Imbalance Handling

- Strategy: `class_weight='balanced'` (inverse class frequency). Applied natively inside each CV fold via model parameter
- Metrics chosen to reflect imbalance: Balanced Accuracy & Precision - Recall

Metric Justification

- Primary metric: Balanced Accuracy: penalises equally all classes regardless of frequency
- High cost of False Negatives (missing a diabetic patient causes irreversible disease progression); FP only triggers a low-cost confirmatory test. Secondary: Precision - Recall

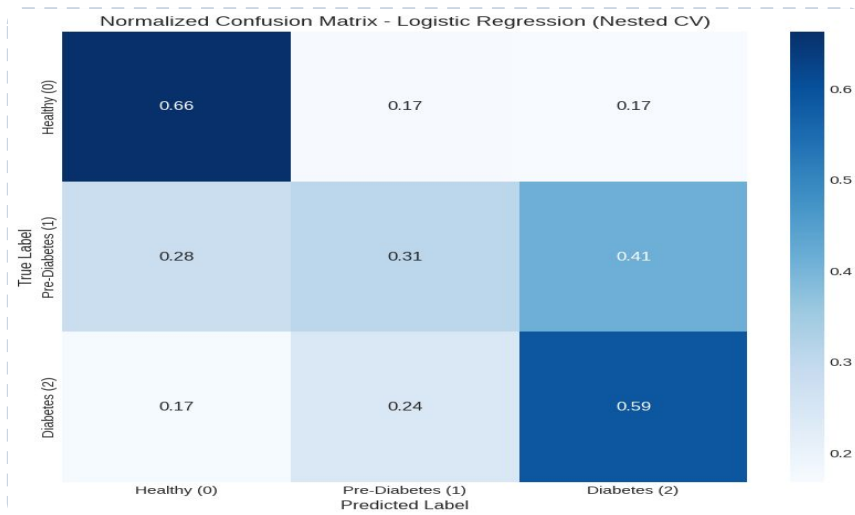
Model Selection & Hyperparameter Optimization



Key Decisions

Candidate models	Random Forest, AdaBoost, XGBoost, Naive Bayes, Logistic Regression: spanning linear, ensemble and probabilistic families
CV strategy	Nested Stratified group k-fold CV: outer 10-fold (model selection) + inner 5-fold (hyperparameter search)
Selection criterion	Balanced Accuracy (outer fold mean $\pm$ std)
Chosen model	Logistic Regression: BA (0.5200 ± 0.0054) and best stability. One of the best Precision-Recall trade-offs
Search strategy	Grid Search (GridSearchCV)
Search space	$C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$; solver $\in \{\text{lbfgs}\}$
Best config	$C=0.1$, solver=lbfgs
Statistical test	Wilcoxon: significant for LR vs RF ($p=0.0020$); non-significant for LR vs XGBoost ($p=0.0645$)

Test-Set Results



Best Hyperparameters: {'model__C': 0.1}

--- Final Model Parameters (Coefficients) ---

	Healthy (Class 0)	Pre-Diabetes (Class 1)	Diabetes (Class 2)
HighBP	-0.182477	-0.002031	0.184508
HighChol	-0.197959	0.088206	0.109754
CholCheck	-0.137264	0.026756	0.110508
BMI	-0.317272	0.103389	0.213884
Smoker	0.006547	-0.005250	-0.001296
Stroke	-0.005040	-0.021201	0.026241
HeartDiseaseorAttack	-0.022624	-0.025977	0.048602
PhysActivity	0.014452	0.002194	-0.016646
Fruits	0.011749	0.001030	-0.012779
Veggies	0.011296	-0.008620	-0.002676
HvyAlcoholConsump	0.074894	0.034256	-0.109150
AnyHealthcare	-0.001666	-0.016030	0.017696
NoDocbcCost	-0.049264	0.061372	-0.012108
GenHlth	-0.336428	0.028329	0.308099
MentHlth	-0.009908	0.041295	-0.031387
PhysHlth	0.031739	-0.006212	-0.025527
DiffWalk	-0.000183	-0.022749	0.022932
Sex	-0.059224	-0.011706	0.070930
Age	-0.314610	0.130256	0.184355
Education	0.035774	-0.035368	-0.000406
Income	0.101779	-0.058405	-0.043374

Key Metrics (Test Set)

0.5200 ± 0.0054

BA (test set)

0.4269 ± 0.0027

F1-macro (CV)

0.31

Recall Pre-Diabetes (CV)

0.333

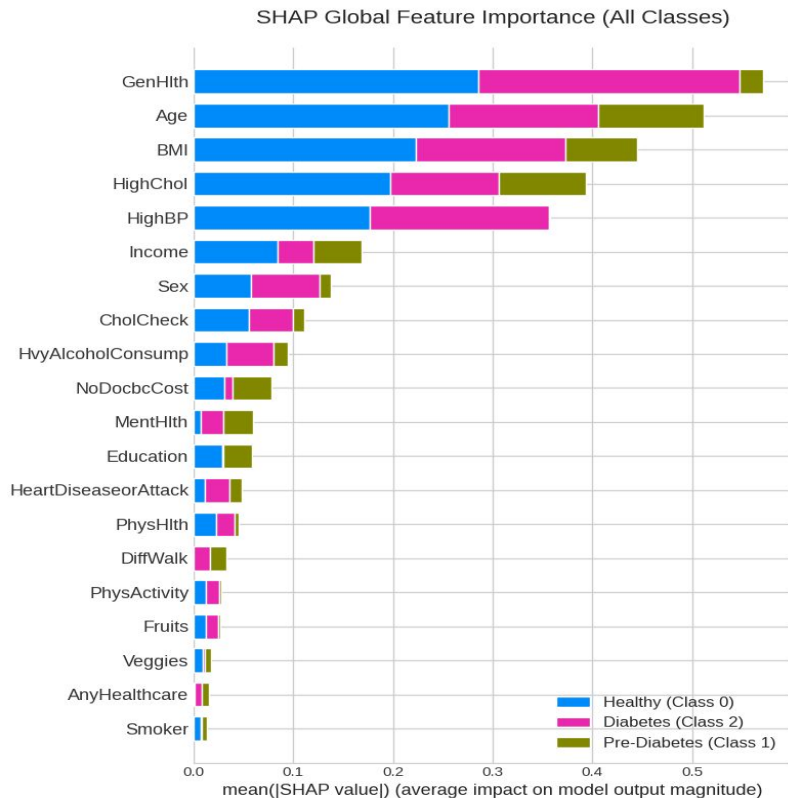
Baseline (random)

PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

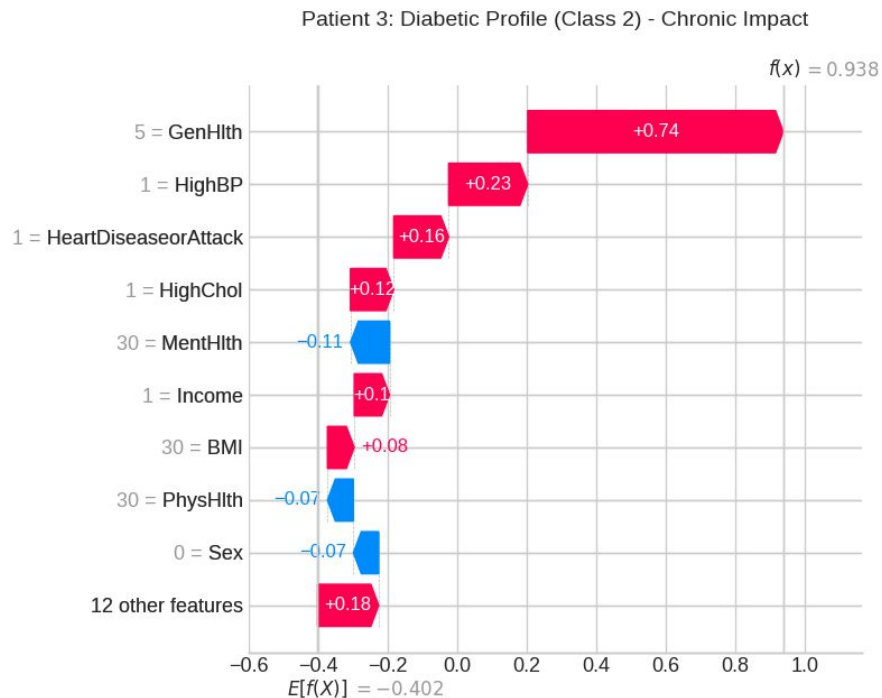
Global Explainability: What Drives the Model Overall?



Interpretation

- XAI Framework:** Post-hoc interpretability via SHAP; game-theoretic Shapley values natively map onto the additive, linear log-odds structure of Logistic Regression.
- Global Core Drivers:** GenHlth, Age, BMI and HighChol dominate the ranking, strongly validating established clinical domain knowledge and epidemiological literature.
- Socio-Economic Proxy:** Income outranks traditional lifestyle factors (PhysActivity, Fruits/Veggies). Structural socio-economic determinants act as stronger global predictors than self-reported behavioral choices.
- Performance Alignment:** SHAP hierarchy strongly supports model metrics; classification relies on a logical cascade of physiological/demographic drivers rather than statistical noise, confirming robust decision boundaries.

Local Explainability: Why This Specific Prediction?

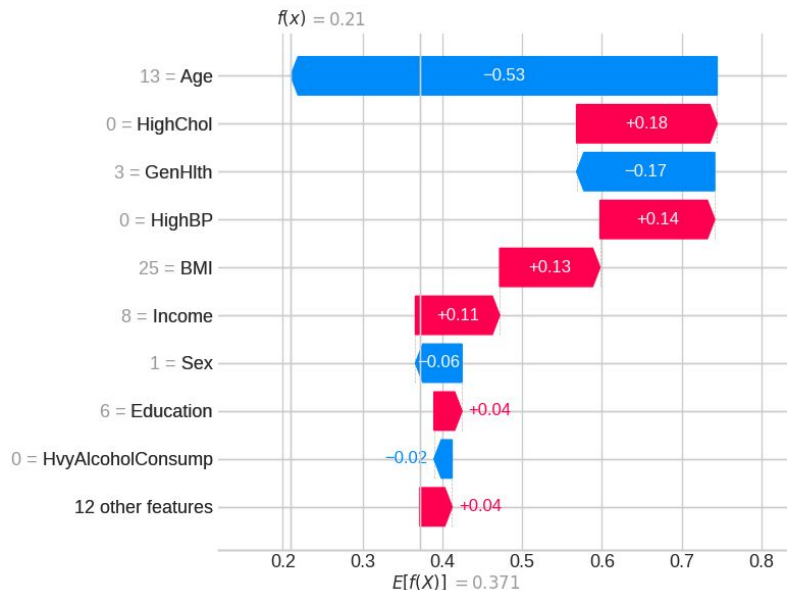


Case Analysis

- **Sample & Ground Truth:** True positive instance of Chronic Diabetes (Class 2). Final prediction scales to $f(x) = 0.938$ against the baseline $E[f(X)] = -0.402$.
- **Predictive Drivers:** Poor perceived health (GenHlth = 5) acts as the decisive trigger (+0.74), heavily compounded by cardiovascular comorbidities (HighBP = 1, HighChol = 1).
- **Global Alignment:** High consistency; the top local drivers (GenHlth, HighBP, HighChol) perfectly mirror the highest-ranked variables from the global macro-analysis.
- **Domain Validation:** Strongly matches epidemiological literature; the co-occurrence of metabolic syndrome markers and poor physical health reflects the classic clinical risk profile.

Error Analysis: What Did the Model Get Wrong and Why?

Patient 4: Diagnostic Error (True risk 2.0, predicted Healthy) - Why did the model fail?



Reverse-Engineering Errors

Systematic Failure Mode: False Negative profile (Patient: Class 2 ground truth, misclassified as Class 0). Highly critical clinical risk.

- **The Masking Mechanism:** Accumulation of linear protective signals mathematically outweighs a massive biological risk factor.
 - **Algebraic Neutralization:** Normal vitals (HighBP=0 (+0.14), HighChol=0 (+0.18), BMI=25 (+0.13)) and high socioeconomic standing (Income=8 (+0.11), Education=6 (+0.04)) provide a combined +0.64 force.
 - **The Blind Spot:** This additive sum completely counterbalances extreme age (Age=13 (-0.53)), causing the final log-odds to land at 0.21 (baseline 0.371).
- Architectural Limit:** Logistic Regression lacks nonlinear interactions, failing on atypical phenotypes (e.g., senile diabetes in lean individuals). Highlight the need for human clinical oversight.

Domain Impact, Limitations & Future Directions

Domain Impact

Practical Use: Non-invasive digital screening and population-level triage tool for early preventative intervention.

Decision Support: Shifts diagnosis from a binary switch to a multi-class risk continuum (Healthy -> Pre-Diabetes -> Diabetes).

Asymmetric Error Costs: High cost asymmetry; False Negatives cause irreversible disease progression, while False Positives only incur low-cost clinical confirmation tests.

Human Oversight: Purely a fast screening decision support tool; requires mandatory physician oversight to validate results and make final diagnostic decisions.

Limitations

Severe Class Imbalance: The strong underrepresentation of Pre-Diabetes and Diabetes classes heavily penalizes multi-class boundary separation, forcing reliance on algorithmic reweighting.

Geographic Representativeness: Dataset is strictly limited to the US population (BRFSS survey); model generalization to other global populations, genetics, and lifestyles remains unverified.

Linearity Constraints: The additive assumption of Logistic Regression cannot capture complex, non-linear risk interactions, causing blind spots in atypical clinical phenotypes.

Future Directions

Global Dataset Expansion: Extend validation to international, multi-ethnic cohorts to verify and retrain the model against diverse genetic backgrounds and cultural lifestyles.

Bio-Clinical Feature Integration: Supplement behavioral survey data with advanced anthropometrics (waist-to-hip ratio) and clinical biomarkers (HbA1c, family medical history).

Clinical Pilot & Deployment: Conduct a real-world deployment study within a primary care setting to gather physician feedback on triage speed, trust, and system usability.

Thank You

Conclusions:

- **The Task:** Multi-class diabetes risk stratification using behavioral and lifestyle survey data.
- **The Pipeline:** Cost-sensitive Logistic Regression validated via robust Nested CV.
- **Key XAI Insight:** SHAP successfully isolated atypical phenotypes masked by linear interaction limits.
- **Next Step:** Integrate biochemical markers (HbA1c) to resolve clinical overlap in pre-diabetes.

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